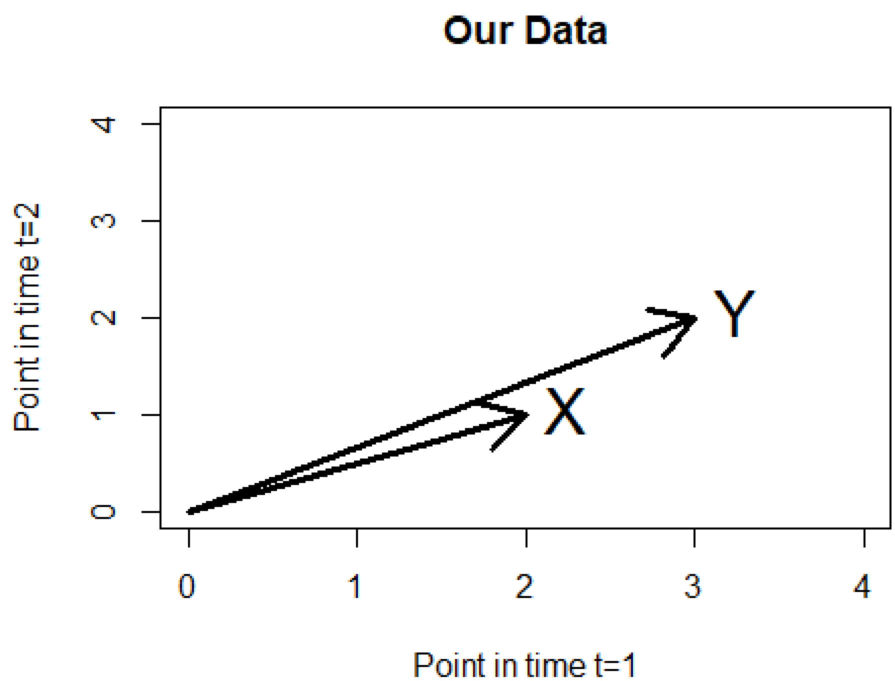
GML_05_02-projection.R

frank

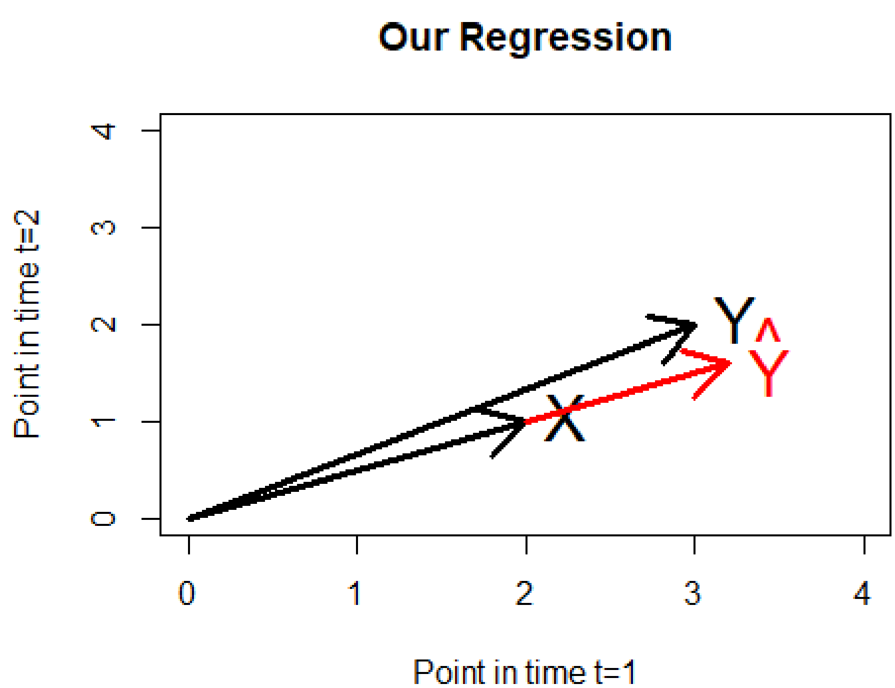
2024-05-06

```
### Willkommen in R!

# lin algebra of lin reg in a nutshell

# Clean previous variables

rm(list=ls(all=TRUE))

#define a Y
Y = matrix(0,2,1)
Y[1,1]=3
Y[2,1]=2
Y

##      [,1]
## [1,]    3
## [2,]    2

X = matrix(0,2,1)
X[1,1]=2
X[2,1]=1
X

##      [,1]
## [1,]    2
## [2,]    1

plot(c(0,4),c(0,4),xlab="Point in time t=1", ylab ="Point in time t=2", type
= "n", main = "Our Data")
arrows(0,0,Y[1,1],Y[2,1],lwd=3)
text(Y[1,1],Y[2,1],"Y",pos=4,cex=2)
arrows(0,0,X[1,1],X[2,1],lwd=3)
text(X[1,1],X[2,1],"X",pos=4,cex=2)
```




```
#do reg without abcissa cause it would be trivial to explain two obs with two regressors X and (1,1)!
#would result in zero residuals!
reg = lm(Y~X-1)
summary(reg)

##
## Call:
## lm(formula = Y ~ X - 1)
##
## Residuals:
##    1    2
## -0.2  0.4
##
## Coefficients:
##   Estimate Std. Error t value Pr(>|t|)
## X      1.6        0.2       8   0.0792 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4472 on 1 degrees of freedom
## Multiple R-squared:  0.9846, Adjusted R-squared:  0.9692
## F-statistic:    64 on 1 and 1 DF,  p-value: 0.07917

#so what's going on here?
x11()
plot(c(0,4),c(0,4),xlab="Point in time t=1", ylab ="Point in time t=2", type = "n", main = "Our Regression")
```

```
arrows(0,0,Y[1,1],Y[2,1],lwd=3)
text(Y[1,1],Y[2,1],"Y",pos=4,cex=2)
arrows(0,0,X[1,1],X[2,1],lwd=3)
text(X[1,1],X[2,1],"X",pos=4,cex=2)

sample_XY = crossprod(X,Y)#scalar product x1y1+x2y2
sample_XY

##      [,1]
## [1,]    8

sample_XX = crossprod(X,X)

b_hat = sample_XY / sample_XX
Y_hat = coef(reg)[1] * X

arrows(X[1,1],X[2,1],Y_hat[1,1],Y_hat[2,1],lwd=3,col="red")
text(Y_hat[1,1],Y_hat[2,1],expression(hat(Y)),pos=4,cex=2,col="red")
```




```
#compare slope b estimators
b_hat#as we have learned

##      [,1]
## [1,]  1.6

coef(reg)[1]#the easy way

##   X
## 1.6
```

```r
#do dumb stuff
reg_dumb = lm(Y~X)
summary(reg_dumb)

##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
## ALL 2 residuals are 0: no residual degrees of freedom!
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1        NaN     NaN      NaN
## X                  1        NaN     NaN      NaN
##
## Residual standard error: NaN on 0 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:    NaN
## F-statistic:   NaN on 1 and 0 DF,  p-value: NA

#and conclude with some reflections
var(X)#note that the squared diffs to mean are divided by n-2=1!

##      [,1]
## [1,]  0.5

#for the following read Assenmacher page 87ff (relation between crossproducts
and moments)
#in case we had an abcissa as in the last regression we could determine the
slope by
cov(X,Y)/var(X)

##      [,1]
## [1,]    1

coef(reg_dumb)[2]

## X
## 1

#but without intercept it is as above and cov/var does not work so easily ...



#and what about the residuals?---------------------------------------------------
--
Xinv = solve(crossprod(X))
resis = Y-X %*% Xinv %*% crossprod(X,Y)

#cross check
residuals(reg)
```

```
##    1    2
## -0.2  0.4
```

```r
resis
```

```
##      [,1]
## [1,] -0.2
## [2,]  0.4
```

```r
#ready to get the std dev!----------------------------------------------

#go for sigma-hat.................................
n=length(X)
k=dim(X)[2]
n
```

```
## [1] 2
```

```r
k #number of regressors
```

```
## [1] 1
```

```r
sigma_hat = sqrt(sum(resis^2)/(n-k))
#note division by degrees of freedom makes this an unbiased estimator! in
contrast ML esti!

#cross check
summary(reg)
```

```
##
## Call:
## lm(formula = Y ~ X - 1)
##
## Residuals:
##    1    2
## -0.2  0.4
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```

```r
sigma_hat
```

```
## [1] 0.4472136
```

```r
#and std err of slope.............................
std_error = sqrt(diag(Xinv))*sigma_hat
```

```r
#cross check
summary(reg)

##
## Call:
## lm(formula = Y ~ X - 1)
##
## Residuals:
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##
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std_error

## [1] 0.2

#and a final word:
#demean X towards Xd
Xd = X-mean(X)
crossprod(Xd,Xd)

##      [,1]
## [1,]  0.5

var(X)#now it fits

##      [,1]
## [1,]  0.5

Yd=Y-mean(Y)
crossprod(Yd,Xd)

##      [,1]
## [1,]  0.5

cov(Y,X)#now it fits

##      [,1]
## [1,]  0.5

#end
```